

KARTHIK PASULA

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Nandyal, India

PROFESSIONAL SUMMARY

Associate Software Engineer / Junior Backend Developer — Backend-focused CS graduate with hands-on experience building production-grade REST APIs, microservices, and containerised deployments. Independently designed and deployed a FastAPI authentication system with JWT, RBAC, PostgreSQL, and Docker (GitHub). Proficient in SQLAlchemy ORM, pytest, Bash scripting, Linux, and Agile practices. Strong problem-solving and communication skills.

TECHNICAL SKILLS

Languages : Python, SQL, Bash Scripting

Backend : FastAPI, REST APIs, JWT Authentication, Microservices, RBAC, Pydantic, bcrypt

Database : PostgreSQL, SQLAlchemy ORM, RDBMS, SQL Queries & Optimization

DevOps & Tools : Git, GitHub, Docker, Containerization Linux, VS Code, CI/CD Basics, Render

Protocols : HTTP / HTTPS , JSON, OpenAPI / Swagger

Concepts : API Design & Integration, Authentication & Authorization, CRUD Operations, OOP

PROJECTS

FastAPI Authentication API | Python, FastAPI, PostgreSQL, SQLAlchemy, JWT, Docker, Pydantic, bcrypt

- Built 15+ RESTful API endpoints using FastAPI and PostgreSQL following clean architecture and separation-of-concerns principles.
- Implemented JWT-based authentication and Role-Based Access Control (RBAC) for secure access management.
- Reduced DB query response time by ~35% via SQLAlchemy ORM optimisation and indexed queries.
- Applied bcrypt password hashing, secure session management, and Pydantic input validation to ensure data integrity.
- Structured modular backend architecture to ensure scalability and maintainability.

GitHub: [Pasulakarthik/new_api](#)

API Link: [Home](#)

System Monitoring Tool | Bash, Linux

GitHub: [Pasulakarthik/Bash-Projects](#)

- Developed a Bash script to monitor CPU, memory, and disk utilisation in real time with configurable alert thresholds.
- Implemented configurable threshold alerts and automated logging for system health tracking.

Docker Deployment Automation | Bash, Docker

- Automated complete Docker pipeline (image pull, restart, port mapping, env config) using Bash scripts.
- Integrated image pull, container restart, port mapping, and environment configuration.

Log Analyzer | Bash, Linux

- Parsed web server logs to extract traffic insights, error statistics, and usage patterns.
- Automated log analysis for performance monitoring and troubleshooting workflow.

Frontend Clones - Bigbasket & Flipkart | HTML, CSS

Webpage Link: [Bigbasket](#) [flipkart](#)

- Built responsive homepage clones of Flipkart and Bigbasket using HTML and CSS.
- Recreated navigation bars, product sections, banners and layouts with responsive design for multiple screen sizes.

EDUCATION

Bachelor of Computer Science (B.Sc. Cs)

[Sri Ramakrishna Degree College (Autonomous)] CGPA: 8.00

Nandyal
2023 -2026

CERTIFICATION

- Python – Reliance Foundation Skilling Academy (2025).
- Communication Skills – APSSDC (2024).

LANGUAGES KNOWN

- English–Speak and Write.
- Telugu–Speak and Write.